

Axiom of Choice and Its Equivalence

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This is a summary about proof of AC, HP, ZL, and WT. Our strategy is circular argument.

Dependency. Logic, Code, Set, Function, Order

Notation.

- $\mathcal{C}(A, \leq) := \{\emptyset \neq T \subseteq A : \leq|_T \text{ is Total}\}$
- $\text{Max}(A, \leq) := \{a \in A : \forall x \in A, a \not\leq x\}$

Axiom of Choice (AC). In $\text{Set}[I]$ A

$$\forall i \in I, A_i \neq \emptyset \Rightarrow \prod_{i \in I} A_i \neq \emptyset.$$

Proof. $\exists \text{Well} \leq_{\text{on}} \bigcup_{i \in I} A_i$ (WT) $\Rightarrow (\min A_i)_{i \in I} \in \prod_{i \in I} A_i$.

Hausdorff Principle (HP). $\forall (A \neq \emptyset, \leq), \text{Max}(\mathcal{C}, \subseteq) \neq \emptyset$

Proof. $f :=$

$$T \rightarrow (\exists C \in \mathcal{C} \text{ s.t. } T \subset C ? C : T) : \mathcal{C} \rightarrow \mathcal{C}$$

s.t. $T \subseteq f(T)$ (AC) $\Rightarrow (\mathcal{C}, \subseteq)$ s.t.

- $\exists M \in \mathcal{C} : \exists a \in A \Rightarrow \{a\} \in \mathcal{C}$
- $\forall \mathcal{T} \in \mathcal{C}(\mathcal{C}), \sup \mathcal{T} \in \mathcal{C} : \forall (x, y) \in \sup \mathcal{T}^2,$

$$\exists (X, Y) \in \mathcal{T}^2 \text{ s.t. } (x, y) \in (X, Y) \Rightarrow \begin{pmatrix} X \subseteq Y \Rightarrow (x, y) \in Y^2 \\ X \supseteq Y \Rightarrow (x, y) \in X^2 \end{pmatrix}$$

$\Rightarrow \mathcal{M} := \min\{\mathcal{X} \subseteq \mathcal{C} :$

- $M \in \mathcal{X}$
- $C \in \mathcal{X} \Rightarrow f(C) \in \mathcal{X}$

- $\mathcal{T} \in \mathcal{C}(\mathcal{X}, \subseteq) \Rightarrow \sup \mathcal{T} \in \mathcal{X}$

} s.t. $[M,]_{\mathcal{M}} = \mathcal{M}$:

- $M \subseteq M$
- $(C \in \mathcal{M} \Rightarrow f(C) \in \mathcal{M})$ s.t. $M \subseteq C \subseteq f(C)$
- $\mathcal{T} \in \mathcal{C}[M,]_{\mathcal{M}} \Rightarrow M \subseteq \sup \mathcal{T} \in \mathcal{M}$

$\mathcal{M}_{E \in \mathcal{E}} := \{C \in \mathcal{M} : C \subseteq E \text{ or } f(E) \subseteq C\} = \mathcal{M}$:

- $M \subseteq E$
- $\begin{cases} C \subset E \Rightarrow f(C) \subseteq E \\ C = E \Rightarrow f(E) = f(C) \end{cases} \text{ or } f(E) \subseteq C \subseteq f(C)$
- $\mathcal{T} \in \mathcal{C}(\mathcal{M}_E) \Rightarrow \sup \mathcal{T} \in \mathcal{M} \Rightarrow \begin{cases} \exists C \in \mathcal{T} \text{ s.t. } f(E) \subseteq C \subseteq \sup \mathcal{T} \\ \sup \mathcal{T} \subseteq E \end{cases}$

$\mathcal{E} := \{E \in \mathcal{M} : \forall C \in \mathcal{M}, C \subset E \Rightarrow f(C) \subseteq E\} = \mathcal{M} : \forall C \in \mathcal{M},$

- $C \not\subseteq M$
-

$$C \subset E \Rightarrow f(C) \subseteq E$$

$\Rightarrow$

$$C \subset f(E) \Rightarrow \begin{cases} C \subset E \Rightarrow f(C) \subseteq E \subseteq f(E) \\ C = E \Rightarrow f(C) = f(E) \end{cases}$$

- $\mathcal{T} \in \mathcal{C}(\mathcal{E}) \Rightarrow \sup \mathcal{T} \in \mathcal{M} \Rightarrow$

$$C \subset \sup \mathcal{T} \Rightarrow \exists E \in \mathcal{T} \text{ s.t. } \begin{cases} C \subset E \Rightarrow f(C) \subseteq E \subseteq \sup \mathcal{T} \\ C = E \Rightarrow f(C) = f(E) \subseteq \sup \mathcal{T} \end{cases}$$

$\Rightarrow \mathcal{M} \in \mathcal{C}$:

$$\forall (X, Y) \in \mathcal{M}^2, Y \in \mathcal{M}_X \Rightarrow Y \subseteq X \text{ or } X \subseteq f(X) \subseteq Y$$

$\Rightarrow T := \sup \mathcal{M} \in \mathcal{M} \Rightarrow f(T) \in \mathcal{M} \Rightarrow T \in \text{Max}(\mathcal{C}, \subseteq) : f(T) = T$

Zorn's Lemma (ZL). In $(A \neq \emptyset, \leq)$

$$\forall T \in \mathcal{C}, T \text{ has upper bound} \Rightarrow \text{Max } A \neq \emptyset$$

Proof. $\exists T \in \text{Max } \mathcal{C}$ (HP) $\Rightarrow \exists u$ is upper bound of $T \Rightarrow u \in \text{Max } A$:

$$\exists a \in A \text{ s.t. } u < a \Rightarrow T \subset T \cup \{a\} \in \mathcal{C}$$

Well-Ordering Theorem (WT). $\forall A \neq \emptyset, \exists \text{Well} \leq_{\text{on}} A$

Proof. In

- $\mathcal{W} := \{(X, \leq) : X \subseteq A, \leq \text{is Well}\} \neq \emptyset :$

$$\exists a \in A \Rightarrow (\{a\}, a \leq a) \in \mathcal{W}$$

- $(X, \leq_1) \leq (Y, \leq_2) : \begin{cases} X \subseteq Y \\ \leq_1 \subseteq \leq_2 \\ \forall (x, y) \in X \times (Y - X), x \leq_2 y \end{cases}$

$$\forall \mathcal{T} \in \mathcal{C}(\mathcal{W}), \sup \mathcal{T} \in \mathcal{W} : \forall \emptyset \neq X \subseteq \sup \mathcal{T},$$

$$\exists W \in \mathcal{T} \text{ s.t. } X \cap W \neq \emptyset \Rightarrow \min X = \min(X \cap W)$$

$$\Rightarrow \exists X \in \text{Max } \mathcal{W} \text{ (ZL)} \Rightarrow X = A :$$

$$X \subset A \Rightarrow \exists x \notin X \text{ s.t. } X \subset X \cup \{x\} \in \mathcal{W}$$

Reference

[1] Serge Lang “Algebra”

[2] Gerald B. Folland “Real Analysis: Modern Techniques and Their Applications”